Solutions to Quiz 4

Question 1

Find elements p, q in $\{0, \ldots, 6\}$ such that if $a = 1 + 7p + 7^2 q$, then $a^2 - 8$ is divisible by 7^3.

Solution:

We calculate

$$a^2 = (1 + 7p + 7^2 q)^2 = 1 + 7(2p) + 7^2(p^2 + 2q) + 7^3 r$$

for a suitable r. Thus, we need $2p - 1$ to be divisible by 7. This gives $p = 4$. We then calculate

$$a^2 - 8 = 7(2p - 1) + 7^2(p^2 + 2q) + 7^3 r$$

Since $2p - 1 = 2 \cdot 4 - 1 = 7$, we need $1 + 4^2 + 2q$ to be divisible by 7. This means $3 + 2q$ has to be divisible by 7 which is satisfied for $q = 2$.

Thus, we have $a = 1 + 7 \cdot 4 + 7^2 \cdot 2$.

We check

$$a^2 = 1 + 7 \cdot 2 \cdot 4 + 7^2(4^2 + 2 \cdot 2) + 7^3 r = 8 + 7^2(1 + 16 + 4) + 7^3 r \equiv 8 \bmod 7^3$$

Question 2

Show that there is an element a in $\mathbb{Z}_5$ such that $a^3 = 26$ in $\mathbb{Z}_5$.

Solution A:

We note that $26 = 1 + 25$ lies in $U_1(\mathbb{Z}_5) = 1 + 5\mathbb{Z}_5$. Thus, the series

$$\operatorname{plog}(26) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{25^k}{k}$$

converges to some element b in $5\mathbb{Z}_5$. Since 3 is a unit in $\mathbb{Z}_5$, we see that there is an element c in $5\mathbb{Z}_5$ such that $3c = b$. It follows that the series

$$\operatorname{pexp}(c) = \sum_{k=0}^{\infty} \frac{c^k}{k!}$$

converges to some element a in $1 + 5\mathbb{Z}_5$. By standard identities, we see that

$$a^3 = \operatorname{pexp}(3c) = \operatorname{pexp}(b)$$

and

$$\operatorname{pexp}(b) = 26$$

It follows that $a^3 = 26$ in $\mathbb{Z}_5$.

Solution B:

We consider the power series (general Binomial series):

$$(1+x)^\alpha = 1 + \sum_{k=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!} x^k$$

When $\alpha = 1/3$, this becomes

$$(1+x)^{1/3} = 1 + \frac{x}{3} + \sum_{m=1}^{\infty} (-1)^m \frac{\prod_{r=1}^{m}(3r-1)}{m+1!} \left(\frac{x}{3}\right)^{m+1}$$

Since 3 is invertible in $\mathbb{Z}_5$, we see that $v_5(x/3) = v_5(x)$. So, if $v_5(x) > 0$, then the series converges in $\mathbb{Z}_5$.

In particular, we see that $(1+25)^{1/3}$ converges in $\mathbb{Z}_5$. Its limit a satisfies $a^3 = 26$.

Solution C:

We note that if $f(x) = x^3 - 26$, then $f'(x) = 3x^2$. Thus, $f'(1) = 3$ is invertible in $\mathbb{Z}_5$. So we write the Newton-Raphson-Hensel iteration

$$x_0 = 1 \text{ and } x_{n+1} = x_n - \frac{f(x_n)}{3}$$

and note that

$$f(x_{n+1}) = f(x_n) - f'(x_n)\frac{f(x_n)}{3} + g(x_n, x_{n+1}) \left(\frac{f(x_n)}{3}\right)^2$$

for some polynomial $g(x, y)$ with coefficients in $\mathbb{Z}_5$.

By induction, we see that $v_5(x_n - 1) > 0$, so that $v_5(x_n^2 - 1) > 0$. It follows that

$$v_5\left(1 - \frac{f'(x_n)}{3}\right) = v_5\left(1 - x_n^2\right) > 0$$

Since

$$f(x_{n+1}) = \left(1 - \frac{f'(x_n)}{3}\right) f(x_n) + g(x_n, x_{n+1}) \left(\frac{f(x_n)}{3}\right)^2$$

we conclude that $v_5(f(x_{n+1})) > v_5(f(x_n))$ so that (x_n) is a convergent sequence. If a is the limit, we have $v_5(f(a)) = \infty$ so that $f(a) = 0$.